BEST1-related Vitreoretinochoroidopathy (ADVIRC): A Comprehensive Disease Characterization

Disease: BEST1-related Vitreoretinochoroidopathy (Autosomal Dominant Vitreoretinochoroidopathy, ADVIRC) **MONDO ID:** MONDO:0700240 · **OMIM:** 193220 · **Causal gene: BEST1** (bestrophin-1; VMD2) **Category:** Mendelian, autosomal dominant inherited retinal dystrophy

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Summary

Autosomal dominant vitreoretinochoroidopathy (ADVIRC) is a rare Mendelian inherited retinal dystrophy caused by heterozygous mutations in the *BEST1* gene (also called *VMD2*), which encodes bestrophin-1 — a ~580-amino-acid, ~66 kDa pentameric calcium-activated chloride channel located in the basolateral plasma membrane of the retinal pigment epithelium (RPE). It is one of the three classic "bestrophinopathies," alongside Best vitelliform macular dystrophy (BVMD) and autosomal recessive bestrophinopathy (ARB), and it is the rarest of the three. ADVIRC is defined clinically by a pathognomonic 360-degree circumferential band of peripheral retinal hyperpigmentation with a discrete posterior (equatorial) boundary, together with fibrillar vitreous condensation, retinal vascular leakage and neovascularization, choroidal atrophy, and a spectrum of developmental ocular anomalies including nanophthalmos/microcornea, iris dysgenesis, cataract, and angle-closure glaucoma.

What mechanistically distinguishes ADVIRC from the other bestrophinopathies is its molecular origin. The causative *BEST1* variants cluster at exonic splice-regulatory elements and simultaneously produce a missense substitution **and** exon skipping, generating three bestrophin-1 isoforms per heterozygous individual (wild-type, missense, and in-frame deletion). These aberrant isoforms co-assemble into the homopentameric channel and confer a **gain-of-function** effect, in contrast to the loss-of-function or dominant-negative mechanisms that drive BVMD and ARB. The downstream consequences branch into a developmental arm (ocular malformations arising during eye development) and a degenerative arm (progressive RPE dysfunction, photoreceptor loss, blood-retinal barrier breakdown, and choroidal atrophy).

Clinically, ADVIRC is slowly progressive with markedly variable expressivity — even within a single family carrying the same variant — ranging from mild peripheral changes with preserved acuity to severe pan-ocular disease with no light perception. There is no disease-specific or curative therapy; management is supportive, addressing macular edema, neovascularization, glaucoma, cataract, and retinal detachment. Because the mechanism is gain-of-function, conventional gene-supplementation strategies are expected to fail, and allele-specific suppression (e.g., CRISPR/Cas9 editing of the mutant allele) is the leading experimental therapeutic avenue. This report consolidates seven confirmed findings and 52 reviewed papers into a structured characterization across all requested domains.

Key Findings

Finding 1 — ADVIRC is caused by *BEST1* (VMD2) mutations acting through aberrant pre-mRNA splicing

The landmark study establishing the molecular basis of ADVIRC was Yardley et al. (2004). Using genetic linkage followed by sequencing in five ADVIRC families, the authors identified three pathogenic *VMD2*/*BEST1* sequence alterations (reported variants include V86M, Y227N, and V235A), each affecting exonic splice enhancer/silencer sites. A minigene splicing assay demonstrated that each mutation simultaneously produces a missense substitution **and** exon skipping. The verbatim conclusion was that *"All sequences showed simultaneous missense substitutions and exon skipping"* PMID: 15452077. The consequence is that each heterozygous individual expresses three distinct bestrophin isoforms: the wild-type protein, a full-length protein carrying the missense substitution, and an in-frame deletion product from skipping the affected exon.

This paper also pinpointed the gene product and its localization: *"VMD2 encodes bestrophin, a transmembrane protein located at the basolateral membrane of the RPE, that is also mutated in Best macular dystrophy"* PMID: 15452077. The study additionally linked ADVIRC-associated splice mutations to nanophthalmos, connecting the splicing defect to a developmental ocular phenotype. This establishes the **initiating molecular lesion** of ADVIRC.

- **Gene annotation:** HGNC:1114 (*BEST1*), OMIM gene 607854.
- **Evidence type:** Human clinical/genetic + in vitro minigene assay.

Finding 2 — ADVIRC is associated with gain-of-function in bestrophin-1, unlike other bestrophinopathies

A recent retrospective case series by Ni et al. (2026) compared 8 ADVIRC, 10 BVMD, and 10 control eyes using quantitative imaging. The ADVIRC outer-segment layer was significantly thinner than in BVMD at all foveal locations, and the ADVIRC photoreceptor layer was thinner than in both BVMD and controls perifoveally (Kruskal–Wallis / Mann–Whitney tests). A companion systematic review documented increased myopia prevalence in ADVIRC (median axial length 24.7 mm, spherical refraction −0.63 D) relative to BVMD and controls. The authors framed the mechanism explicitly: *"Autosomal dominant vitreoretinochoroidopathy (ADVIRC) is associated with gain-of-function in bestrophin-1 instead of loss-of-function or dominant-negative mechanisms in other bestrophinopathies"* PMID: 41991505, and connected that mechanism to phenotype: *"This study found that gain-of-function in ADVIRC may cause increased prevalence of myopia, PR layer thinning, and outer segment shortening"* PMID: 41991505.

This gain-of-function framing is mechanistically pivotal: it explains why ADVIRC presents with developmental anomalies and a distinct degenerative profile, and it has direct therapeutic implications (gene supplementation would not overcome a toxic gain-of-function allele).

- **Evidence type:** Human clinical imaging cohort + systematic review.

Finding 3 — Pathognomonic 360-degree circumferential peripheral pigmentary band with a discrete posterior boundary

Multiple independent case series describe the defining ophthalmoscopic sign of ADVIRC. Blair et al. (1984) described *"peripheral, coarse pigmentary degeneration of the fundus for 360 degrees, with a relatively discrete posterior border in the equatorial region (this finding may be pathognomonic)"* PMID: 6689931, together with punctate yellow-white retinal opacities, vascular abnormalities, blood-retinal barrier breakdown, retinal neovascularization, vitreous abnormalities, and choroidal atrophy.

Roider et al. (1997), studying an independent pedigree (7/15 family members affected), confirmed: *"The main clinical findings were peripheral pigmentary changes for 360 degrees, with a discrete boundary near the equator"* PMID: 9279944, and documented marked reduction of the EOG light rise (reduced Arden ratio), implicating RPE dysfunction.

Goldberg et al. (1989) provided histopathologic correlation, describing ADVIRC as *"clinically characterized by a slowly progressive or stationary circumferential peripheral pigmentary retinopathy with fibrillar condensation of the vitreous"* PMID: 2516300, with disorganized peripheral retina, focally atrophic RPE, perivascular pigment migration, equatorial multifocal photoreceptor loss, and a preretinal membrane of condensed vitreous and Müller cells.

- **HPO suggestions:** Abnormal retinal pigmentation (HP:0007703), Retinal pigment epithelial atrophy (HP:0007894), Peripheral retinal degeneration (HP:0007769), Vitreous floaters / abnormal vitreous morphology (HP:0004327), Chorioretinal atrophy (HP:0000533), Retinal neovascularization (HP:0012808).
- **Evidence type:** Human clinical case series + histopathology.

Finding 4 — BEST1 encodes bestrophin-1, a Ca^{2+}-activated Cl^{-} channel at the RPE basolateral membrane

BEST1 (*VMD2*) encodes bestrophin-1, a ~580-amino-acid, ~66 kDa transmembrane protein. Guziewicz et al. (2007) defined the protein as one that *"encodes bestrophin, a 580-amino acid protein of 66 kDa"* PMID: 17460247 and localized it via immunocytochemistry: *"immunocytochemistry of normal canine retina demonstrated specific localization of protein to the RPE basolateral plasma membranes"* PMID: 17460247.

Functionally, Wang et al. (2026) described bestrophin-1 as *"a calcium-activated chloride channel predominantly expressed in the RPE, where it plays essential roles in ionic homeostasis, transepithelial transport, and regulation of the visual cycle"* PMID: 42222334. Cryo-EM studies confirm that bestrophin-1 assembles as a homopentameric channel with resolved closed and partially open conformations (Owji et al. 2022, PMID: 35789156; Kim et al. 2025, PMID: 40043778). The pentameric architecture is mechanistically important for ADVIRC, because it allows mutant and wild-type subunits to co-assemble.

- **GO suggestions:** chloride channel activity (GO:0005254), calcium-activated chloride channel activity (GO:0005229), chloride transmembrane transport (GO:1902476), basolateral plasma membrane (GO:0016323).
- **CHEBI:** chloride (CHEBI:17996), calcium(2+) (CHEBI:29108).
- **Evidence type:** Model organism (canine) immunocytochemistry + human/structural biology + review.

Finding 5 — ADVIRC causes developmental anterior-segment/ocular anomalies with marked variable expressivity

Vincent et al. (2011) studied a family with the heterozygous *BEST1* variant p.V86M (c.256G>A). Among six subjects, the phenotype spanned: *"angle closure glaucoma (n = 2), microcornea with shallow anterior chamber (n = 1), iris dysgenesis (n = 2), cataracts (n = 4), classical peripheral concentric band of retinal hyperpigmentation (n = 5), and optic nerve dysplasia (n = 1)"* PMID: 21072067. Visual acuity ranged from ≥20/25 to no light perception, and full-field ERG from low-normal to non-recordable with age. The authors characterized the disorder as *"a slowly progressive vitreoretinal degeneration that demonstrates marked intra-familial phenotypic variability"* PMID: 21072067.

Variable expressivity within families carrying the same variant is a recurring theme: da Palma et al. (2021) documented it in a three-generation pedigree with c.715G>A/p.V239M PMID: 34746433, and Mainguy et al. (2024) reported the splice variant c.1101-1G>T with a distinctive ADVIRC phenotype PMID: 38957071. This finding supports the view that ADVIRC is a **pan-ocular developmental and degenerative** condition, not merely a peripheral retinopathy.

- **HPO suggestions:** Glaucoma / Angle-closure glaucoma (HP:0000501 / HP:0012109), Microcornea (HP:0000482), Shallow anterior chamber (HP:0000594), Iris hypoplasia / coloboma (HP:0000526), Cataract (HP:0000518), Optic nerve dysplasia (HP:0001093), Myopia (HP:0000545), Nanophthalmos / Microphthalmia (HP:0000568).
- **Evidence type:** Human clinical/genetic family studies.

Finding 6 — ADVIRC is a rare autosomal dominant bestrophinopathy and a minority of BEST1-related retinal dystrophy

A population registry study by Sæther et al. (2026) of 60 *BEST1*-related retinal dystrophy (BEST1-RD) patients in Southern and Eastern Norway found: *"Forty-one patients were diagnosed with Best's vitelliform macular dystrophy (BVMD), 15 with autosomal recessive bestrophinopathy (ARB) and 4 with autosomal dominant vitreoretinochoroidopathy (ADVIRC)"* PMID: 41627933. ADVIRC was thus the rarest of the three classic bestrophinopathies (4/60; ~6.7%). The same study estimated overall BEST1-RD prevalence: *"The prevalence of BEST1-RD in Southern and Eastern Norway was between 1:64 600 and 1:43 700"* PMID: 41627933, noting *BEST1* as the third leading cause of inherited retinal dystrophy in Norway.

Inheritance is autosomal dominant (Blair 1984 PMID: 6689931; Yardley 2004 PMID: 15452077), contrasting with the autosomal recessive inheritance of ARB.

- **Evidence type:** Population registry / epidemiology.

Finding 7 — Integrated mechanism: splice-altering *BEST1* mutations produce a gain-of-function RPE channel that drives both developmental anomalies and progressive retinopathy

Synthesizing the above, the causal chain runs: (1) a heterozygous *BEST1* exonic splice-site variant (e.g., c.256G>A/p.V86M) causes simultaneous missense substitution and exon skipping, generating three bestrophin-1 isoforms (Yardley 2004 minigene assay, PMID: 15452077); (2) the aberrant isoforms co-assemble into the homopentameric Ca^{2+}-activated Cl^{-} channel (cryo-EM pentamer, Owji 2022 PMID: 35789156); (3) conferring a gain-of-function effect (Ni 2026, PMID: 41991505); which (4) branches into a **developmental arm** (nanophthalmos/myopia, microcornea, iris dysgenesis, cataract, angle-closure glaucoma — Vincent 2011 PMID: 21072067) and a **degenerative arm** (RPE anion/Ca^{2+}/fluid-transport dysfunction — iPSC-RPE models Singh 2013 PMID: 23139242, Li 2017 PMID: 29063836) producing a reduced EOG Arden ratio (Roider 1997 PMID: 9279944), the 360-degree peripheral pigmentary band with equatorial photoreceptor loss (Goldberg 1989 PMID: 2516300), blood-retinal barrier breakdown, neovascularization, macular edema, and choroidal atrophy (Blair 1984 PMID: 6689931). There is no environmental or infectious trigger; disease is germline and autosomal dominant.

Section-by-Section Report

1. Disease Information

ADVIRC is a rare, slowly progressive, autosomal dominant inherited retinal dystrophy affecting the full thickness of the eye — vitreous, retina, RPE, and choroid — hence "vitreoretinochoroidopathy." Its causal gene is **BEST1** (bestrophin-1). The clinical hallmark is a circumferential 360-degree peripheral band of pigmentary retinopathy with a sharply demarcated posterior (equatorial) boundary, accompanied by fibrillar vitreous condensation, retinal vascular leakage, neovascularization, cystoid macular edema, choroidal atrophy, and developmental ocular anomalies.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0700240
OMIM	193220 (Vitreoretinochoroidopathy, VRCP / ADVIRC)
Gene (OMIM)	*BEST1* 607854
HGNC	HGNC:1114 (*BEST1*)
Synonyms	ADVIRC; VRCP; vitreoretinochoroidopathy, autosomal dominant; BEST1-related vitreoretinochoroidopathy
ICD-11	9B70–9B71 range (hereditary retinal dystrophy, closest match)
MeSH	Closest: Retinal Dystrophies / Eye Diseases, Hereditary

Information is derived from **aggregated disease-level resources** (OMIM, Orphanet) and from **individual-patient** case series and family studies in the primary literature; there is no large EHR-derived dataset for this ultra-rare disease.

2. Etiology

Causal factor: Monogenic — heterozygous germline mutations in *BEST1* that disrupt exonic splice-regulatory elements, producing simultaneous missense substitution and exon skipping (Yardley 2004, PMID: 15452077). No environmental or infectious cause is known.

Genetic risk factors: The causal variants themselves are the risk factor. Reported ADVIRC variants include V86M (c.256G>A), Y227N, V235A, p.V239M (c.715G>A), p.T277M (c.830C>T), and the splice variant c.1101-1G>T. These affect exonic splice enhancer/silencer sites.

Environmental risk / protective factors: None established. Because the disease is highly penetrant Mendelian with no known modifiers, environmental contributions are not a described feature. No protective alleles or lifestyle protective factors are documented.

Gene–environment interactions: None documented. The variable expressivity seen within families (same variant, different severity) suggests the influence of stochastic developmental factors or unidentified genetic modifiers rather than environment.

3. Phenotypes

Phenotype	Type	Onset	Severity / Progression	Frequency	Suggested HPO
360° circumferential peripheral pigmentary band, discrete equatorial border	Clinical sign (pathognomonic)	Congenital/ early, progressive	Slowly progressive–stationary	Near-universal	HP:0007703, HP:0007769
Fibrillar vitreous condensation	Clinical sign	Early	Stable–progressive	Common	HP:0004327
Reduced visual acuity / night blindness	Symptom	Childhood–adult	Variable (20/25 to NLP)	Variable	HP:0000572, HP:0000662
Retinal vascular leakage / blood-retinal barrier breakdown	Clinical sign	Variable	Progressive	Common	HP:0000608
Retinal neovascularization	Clinical sign	Later	Progressive	Subset	HP:0012808
Cystoid macular edema	Clinical sign	Variable	Episodic/ progressive	Subset	HP:0011505
Choroidal atrophy	Clinical sign	Progressive	Progressive	Common	HP:0000533
Myopia / increased axial length	Clinical sign	Early	Stable	Increased vs BVMD	HP:0000545
Nanophthalmos / microcornea	Developmental	Congenital	Stable	Subset	HP:0000568, HP:0000482
Iris dysgenesis	Developmental	Congenital	Stable	Subset	HP:0000526
Cataract	Clinical sign	Variable	Progressive	Frequent (4/6 in V86M family)	HP:0000518
Angle-closure glaucoma	Clinical sign	Adult		Subset	HP:0012109

Phenotype	Type	Onset	Severity / Progression	Frequency	Suggested HPO
			Progressive, vision-threatening		
Optic nerve dysplasia	Developmental	Congenital	Stable	Rare	HP:0001093
Reduced EOG Arden ratio	Laboratory/ electrophysiology	Any	Stable	Characteristic	—

Quality of life impact: ADVIRC causes progressive peripheral and (variably) central visual field loss, night blindness, and in severe cases blindness (no light perception), with substantial impact on mobility, driving, and independent daily functioning. Angle-closure glaucoma and cataract add acute vision-threatening and surgical burdens. No disease-specific EQ-5D/SF-36 data exist; impact is inferred from documented visual acuity ranges and progression (Vincent 2011, PMID: 21072067).

4. Genetic / Molecular Information

Causal gene: *BEST1* (HGNC:1114; OMIM 607854; chromosome 11q12.3), encoding bestrophin-1.

Pathogenic variants (ADVIRC-specific):

Variant (cDNA)	Protein	Type	Mechanism note
c.256G>A	p.V86M	Missense + splice	Classic ADVIRC; exonic splice element (Yardley 2004, Vincent 2011)
—	p.Y227N	Missense + splice	Yardley 2004
—	p.V235A	Missense + splice	Yardley 2004
c.715G>A	p.V239M	Missense + splice	da Palma 2021
c.830C>T	p.T277M	Missense	Atypical/posterior ADVIRC (P 36712704)
c.1101-1G>T	splice acceptor	Splice	Mainguy 2024

Variant classification: Pathogenic / likely pathogenic per ACMG/AMP (co-segregation, functional splicing assay, recurrence). **Variant type:** predominantly missense variants that also disrupt splicing (dual consequence), plus canonical splice-site variants. **Functional consequence:** gain-of-function (Ni 2026, PMID: 41991505) — distinct from the loss-of-function (ARB) and dominant-negative (BVMD) mechanisms of other bestrophinopathies. **Origin:** germline. **Allele frequency:** these are private/rare disease variants, essentially absent or ultra-rare in gnomAD.

Modifier genes: None confirmed; variable expressivity suggests undiscovered modifiers or stochastic factors.

Epigenetic information / chromosomal abnormalities: None reported; ADVIRC is a point-mutation/splice disorder without large structural rearrangements.

5. Environmental Information

No environmental, lifestyle, toxic, or infectious factors are implicated in ADVIRC. It is a purely Mendelian germline disorder. This section is **not applicable** beyond noting the absence of such contributions.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **heterozygous *BEST1* exonic splice-regulatory variant** (e.g., c.256G>A/p.V86M) **results in** disruption of an exonic splice enhancer/silencer (Yardley 2004 — *demonstrated*).
2. This **leads to** simultaneous missense substitution **and** exon skipping during pre-mRNA splicing, **producing** three bestrophin-1 isoforms per cell: wild-type, missense, and in-frame-deletion (Yardley 2004 minigene assay — *demonstrated in vitro*).
3. The aberrant isoforms **co-assemble** into the **homopentameric** Ca^{2+}-activated Cl^- channel (cryo-EM pentamer; Owji 2022 — *structural inference applied to ADVIRC*).
4. The mixed-subunit channel **confers a gain-of-function** effect on RPE anion/ion handling (Ni 2026 — *inferred from phenotype and comparative imaging*).
5. **Branch A — developmental arm:** altered RPE/ocular ion and fluid handling during eye development **leads to** nanophthalmos/myopia, microcornea, shallow anterior chamber, iris dysgenesis, cataract, optic nerve dysplasia, and predisposition to **angle-closure glaucoma** (Vincent 2011).
6. **Branch B — degenerative arm:** chronic RPE dysfunction (impaired Ca^{2+} signaling, anion transport, and fluid flux; impaired photoreceptor-outer-segment handling) **results in** RPE atrophy, equatorial photoreceptor loss, a reduced EOG light rise (lowered Arden ratio), the pathognomonic 360° pigmentary band, blood-retinal barrier breakdown, retinal neovascularization, cystoid macular edema, and choroidal atrophy (Goldberg 1989; Roider 1997; Blair 1984).

Molecular pathways / biochemical defect: The primary defect is in a **calcium-activated chloride channel** (bestrophin-1) governing RPE ionic homeostasis, transepithelial fluid transport, and the visual cycle (Wang 2026, PMID: 42222334). iPSC-RPE models of *BEST1* disease demonstrate disrupted fluid flux, delayed rhodopsin/outer-segment degradation, altered calcium responses, and increased oxidative stress (Singh 2013, PMID: 23139242; Li 2017, PMID: 29063836). Note: these functional models are largely of BVMD/ARB-type mutations; direct ADVIRC-isoform functional assays remain limited.

Cellular processes: RPE dysfunction, impaired phagocytosis/outer-segment turnover, oxidative stress, and secondary photoreceptor degeneration.

Tissue damage mechanisms: RPE atrophy, blood-retinal barrier breakdown with vascular leakage and neovascularization, choroidal atrophy, and vitreous fibrillar condensation.

- **GO (biological process) suggestions:** chloride transmembrane transport (GO:1902476), regulation of intracellular pH/ion homeostasis (GO:0006885), visual perception (GO:0007601), retinal pigment epithelium development (GO:0003406), response to oxidative stress (GO:0006979).
- **GO (cellular component):** basolateral plasma membrane (GO:0016323), endoplasmic reticulum (GO:0005783).
- **CL (cell type) suggestions:** retinal pigment epithelial cell (CL:0002586), retinal photoreceptor cell (rod CL:0000604, cone CL:0000573), Müller cell (CL:0000636).

7. Anatomical Structures Affected

- **Primary organ:** eye (UBERON:0000970), specifically the retina (UBERON:0000966), RPE (UBERON:0001782), choroid (UBERON:0001776), and vitreous body (UBERON:0001796).
- **Secondary involvement:** anterior segment — cornea (UBERON:0000964), iris (UBERON:0001769), lens (UBERON:0000965), anterior chamber/trabecular meshwork (angle-closure glaucoma); optic nerve (UBERON:0000941).
- **Body system:** visual/sensory system (nervous system, special sense organ).
- **Tissue/cell level:** epithelial (RPE), neural (photoreceptors, Müller glia), vascular endothelium (retinal vessels).
- **Subcellular level:** basolateral plasma membrane of RPE (GO:0016323); endoplasmic reticulum calcium stores implicated in *BEST1* function (Singh 2013).
- **Lateralization: bilateral**, generally symmetric; the peripheral band is circumferential (360°).

8. Temporal Development

- **Onset:** Congenital developmental anomalies (nanophthalmos, microcornea, iris dysgenesis) are present from birth; the pigmentary retinopathy and degenerative features are early-onset and progressive, often recognized in childhood to adulthood.
- **Onset pattern:** chronic, insidious.
- **Progression:** slowly progressive to stationary (Goldberg 1989, PMID: 2516300); electrophysiology (ffERG) may decline from low-normal to non-recordable with age (Vincent 2011). Course is progressive but slow; disease is **chronic and lifelong**.

- **Variable expressivity:** marked, even within families carrying the same variant (da Palma 2021; Mainguy 2024).
- **Remission:** none; no spontaneous remission. **Critical periods:** eye development (for the congenital anomalies) and ongoing adult life for degenerative progression and glaucoma risk.

9. Inheritance and Population

- **Inheritance:** autosomal dominant (contrast with autosomal recessive ARB).
- **Penetrance:** high/complete for genotype, but **highly variable expressivity**.
- **Epidemiology:** ultra-rare. In a Norwegian registry, ADVIRC accounted for only 4/60 BEST1-RD patients; overall BEST1-RD prevalence was estimated at **1:64,600 to 1:43,700** (Sæther 2026, PMID: 41627933), making ADVIRC a small fraction of that — on the order of 1 in several hundred thousand to low millions.
- **Sex ratio:** no sex predilection reported (autosomal dominant).
- **Founder effects / consanguinity / anticipation / mosaicism:** none established; variants are typically private. No repeat-expansion mechanism, so genetic anticipation is not expected.
- **Carrier frequency:** not applicable (dominant; affected = carriers).

10. Diagnostics

- **Clinical examination:** dilated fundoscopy revealing the pathognomonic 360° peripheral pigmentary band with discrete equatorial boundary; fibrillar vitreous.
- **Imaging:** ultra-widefield fundus photography, fluorescein angiography (vascular leakage, neovascularization, barrier breakdown), OCT (photoreceptor/outer-segment thinning, macular edema), fundus autofluorescence.
- **Electrophysiology: Electrooculography (EOG)** showing reduced Arden ratio (reduced light rise) reflecting RPE dysfunction (Roider 1997, PMID: 9279944); full-field ERG ranging from low-normal to non-recordable with age (note: EOG can occasionally be normal in bestrophinopathies — PMID: 28590961).
- **Genetic testing (confirmatory):** targeted single-gene *BEST1* sequencing, inherited-retinal-dystrophy gene panels, or WES/WGS. Because pathogenic ADVIRC variants act on splicing, functional/minigene confirmation or in silico splice prediction strengthens classification.

- **Histopathology (research/autopsy):** disorganized peripheral retina, focal RPE atrophy, perivascular pigment migration, equatorial photoreceptor loss, preretinal Müller/vitreous membrane (Goldberg 1989).
- **Differential diagnosis:** familial exudative vitreoretinopathy, retinitis pigmentosa, other bestrophinopathies (BVMD, ARB), Wagner syndrome, and retinal vasculopathies. Distinguishing feature: the discrete 360° equatorial pigmentary band and *BEST1* genotype.

11. Outcome / Prognosis

- **Survival/mortality:** ADVIRC is **not life-limiting**; it is an ocular disorder. No disease-specific mortality.
- **Morbidity:** significant visual morbidity — progressive field loss, variable central vision loss (20/25 to no light perception), night blindness, and risk of blindness; secondary complications include angle-closure glaucoma, cataract, cystoid macular edema, neovascularization, and (rarely) retinal detachment.
- **Prognostic factors:** severity of developmental anomalies, presence of glaucoma/neovascularization, age (ffERG declines with age), and specific variant — though expressivity is highly variable.
- **Disease course:** slowly progressive, lifelong; bestrophinopathies generally show slow BCVA decline in the absence of CNV/glaucoma complications (natural-history data for the broader bestrophinopathy group, PMID: 42559932).

12. Treatment

There is **no curative or disease-specific therapy**. Management is supportive and complication-directed:

- **Pharmacotherapy / interventional:** anti-VEGF intravitreal injections for choroidal neovascularization/macular edema; IOP-lowering medications and laser peripheral iridotomy/trabeculectomy for angle-closure glaucoma (as applied in bestrophinopathy cohorts, e.g., PMID: 39048936).
- **Surgical:** cataract extraction; glaucoma surgery; vitrectomy/retinal detachment repair as needed.
- **Supportive/rehabilitative:** low-vision aids, occupational support, and genetic counseling.
- **Experimental / advanced therapeutics:** Because ADVIRC is **gain-of-function**, simple gene-supplementation (adding wild-type *BEST1*) is unlikely to help. The rational strategy is **allele-specific suppression** of the mutant transcript. Proof-of-concept exists: allele-specific

CRISPR/Cas9 editing in hiPSC-RPE restored BEST1 localization, protein stability, and anion transport for selected dominant *BEST1* mutations (PMID: 42010707). AAV-mediated *BEST1* delivery to RPE has been demonstrated in canine models (serotype-dependent; rAAV2/2 safe and RPE-specific, PMID: 24143172), and broader gene-therapy strategies for bestrophinopathies are under review (PMID: 41096689).

- **NCIT term suggestions:** Anti-VEGF Therapy, Laser Therapy (NCIT:C15262), Cataract Surgery, Gene Therapy (NCIT:C15431), Genetic Counseling (NCIT:C15417), Supportive Care (NCIT:C15227).

13. Prevention

- **Primary prevention:** not possible (germline Mendelian). **Genetic counseling** is central: autosomal dominant inheritance confers 50% transmission risk to offspring. **Preimplantation genetic diagnosis (PGD)** and **prenatal testing** are options once the family variant is known.
- **Secondary prevention:** early ophthalmologic surveillance in at-risk family members (cascade testing) to detect and treat angle-closure glaucoma, cataract, macular edema, and neovascularization early.
- **Tertiary prevention:** regular monitoring and timely intervention to preserve remaining vision (IOP control, anti-VEGF, surgery).
- **Immunization / public health / environmental interventions:** not applicable.

14. Other Species / Natural Disease

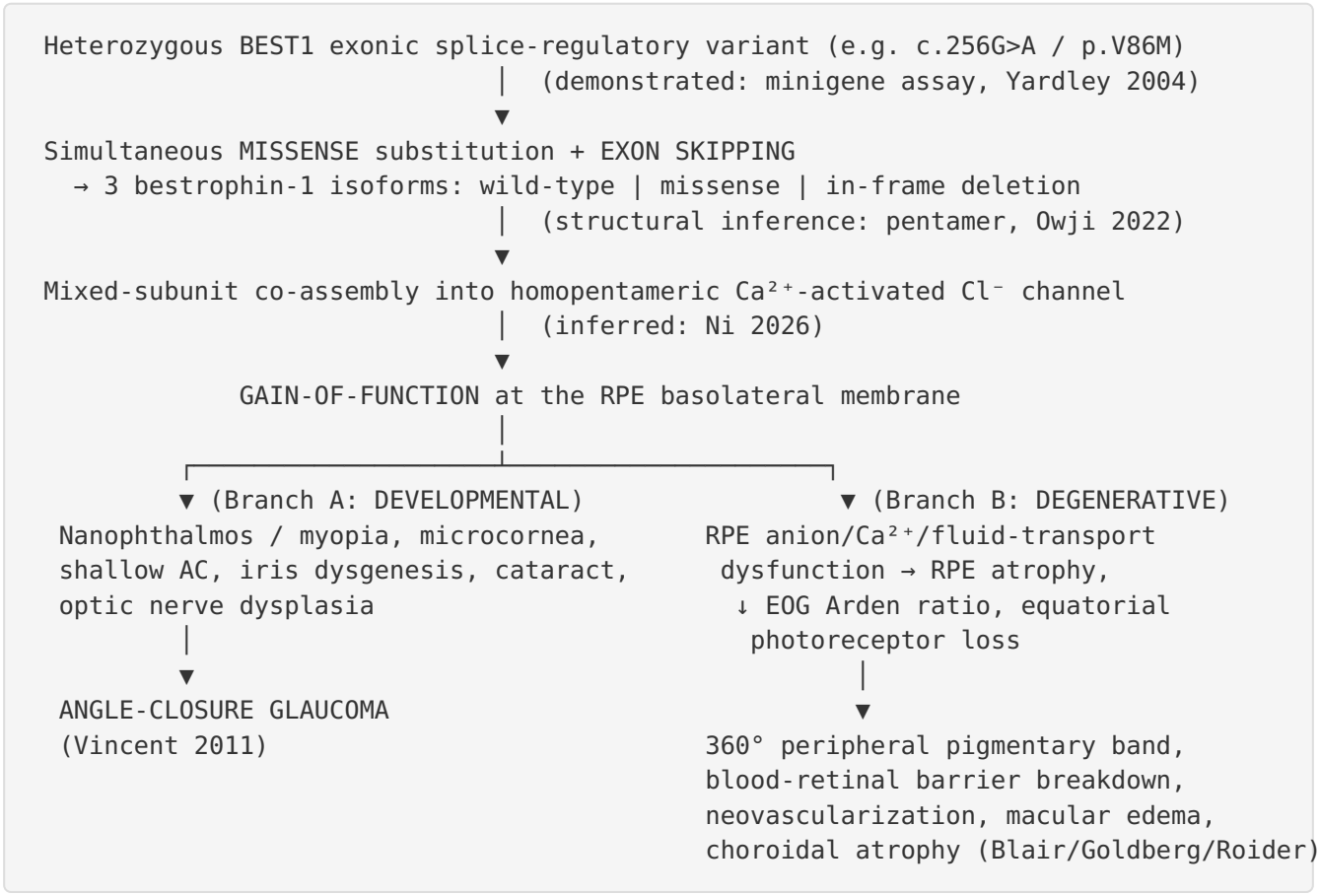
- **Orthologous gene / natural models:** *BEST1* has a canine ortholog, and naturally occurring **canine multifocal retinopathy (cmr)** is caused by *BEST1* mutations — a validated large-animal model of human bestrophinopathy (Guziewicz 2007, PMID: 17460247; cmr3, PMID: 21197113). Canine bestrophinopathy recapitulates the focal/multifocal retina–RPE separations and underdeveloped RPE apical domain seen in human disease (PMID: 29721958).
- **Taxonomy:** *Canis lupus familiaris* (NCBI:txid9615); affected breeds include mastiff-related breeds (cmr1), Coton de Tulear (cmr2), and Lapponian herder (cmr3).
- **Comparative biology:** cmr shares clinical and molecular features with human bestrophinopathies but reflects loss-of-function/mislocalization mechanisms — ADVIRC's specific gain-of-function splicing mechanism is **not** directly modeled by existing cmr dogs.
- **Zoonotic potential:** none (genetic disease).

15. Model Organisms

Model	Type	Use / Recapitulation	Reference
hiPSC-derived RPE (patient-specific)	In vitro / cellular	Disrupted fluid flux, delayed POS/ rhodopsin degradation, altered Ca^{2+} responses, oxidative stress	PMID: 23139242, PMID: 29063836
Canine multifocal retinopathy (cmr)	Mammalian (dog)	Natural *BEST1* model; retina–RPE separation, underdeveloped RPE apical domain	PMID: 17460247, PMID: 29721958
Zebrafish *best1*	Vertebrate	Mutant mislocalization and retinal-integrity effects	PMID: 41456629
MDCK II polarized epithelial cells	In vitro	Channel localization/function assays for *BEST1* mutants	PMID: 41456629

Limitation of models: No existing animal model carries an ADVIRC splicing/gain-of-function variant; most functional and animal work models BVMD/ARB (loss-of-function/dominant-negative) mechanisms. This is a key gap for ADVIRC-specific preclinical therapeutic testing.

Mechanistic Model / Interpretation



The defining conceptual point is that ADVIRC sits apart from the other two bestrophinopathies not by which gene is mutated but by **how** the channel is altered. BVMD arises from dominant-negative mutant subunits poisoning the pentamer; ARB arises from biallelic loss-of-function; ADVIRC arises from splice-altering variants that create a **gain-of-function** channel. This single mechanistic distinction accounts for ADVIRC's unique combination of **congenital developmental anomalies** (which the degenerative-only bestrophinopathies largely lack) plus a **progressive peripheral degeneration** with its signature circumferential band.

Comparison of the three classic bestrophinopathies

Feature	BVMD	ARB	ADVIRC
Inheritance	Autosomal dominant	Autosomal recessive	**Autosomal dominant**
Mechanism	Dominant-negative	Loss-of-function (biallelic)	**Gain-of-function (splice-altering)**
EOG Arden ratio	Reduced	Reduced	**Reduced**
Hallmark lesion	Central vitelliform ("egg-yolk")	Multifocal subretinal deposits	**360° peripheral pigmentary band**
Developmental anomalies	Rare	Angle-closure, hyperopia	**Prominent (nanophthalmos, iris dysgenesis, etc.)**
Relative frequency (Norway)	41/60	15/60	**4/60 (rarest)**

Evidence Base

PMID	Title (abbrev.)	Role in this report
15452077	Mutations of VMD2 splicing regulators cause nanophthalmos and ADVIRC	**Foundational:** establishes splicing mechanism and causal gene
41991505	Gain of function in BEST1: photoreceptor changes and myopia in ADVIRC	**Foundational:** gain-of-function mechanism + PR/refractive phenotype
21072067	BEST1-related ADVIRC: developmental ocular anomalies	Developmental-arm phenotype; variable expressivity (p.V86M)
6689931	ADVIRC (Blair 1984)	Pathognomonic 360° band; vascular/ vitreous/choroidal features
2516300	Histopathologic study of ADVIRC	Histopathology; circumferential pigmentary retinopathy
9279944	ADVIRC (Roider 1997)	Independent family; reduced EOG Arden ratio
17460247	Bestrophin mutations cause canine multifocal retinopathy	Protein size/localization; canine natural model
42222334	From BEST1 mutations to retinal regeneration	Bestrophin-1 molecular function (CaCC, visual cycle)
35789156	Structures and gating of human bestrophin channels	Homopentameric architecture (co-assembly basis)
40043778	Cryo-EM of mouse BEST1 closed/partially open	Conformational states of the channel
41627933	BEST1-associated dystrophies in Norway	Epidemiology; ADVIRC rarest (4/60); prevalence estimate
34746433	Variable expressivity of ADVIRC (p.V239M)	Intra-familial variability with identical variant
38957071	Variable expressivity, c.1101-1G>T	Novel splice variant; variable phenotype
23139242	iPSC modeling of Best disease	RPE fluid flux, Ca^{2+}, POS handling (degenerative arm)
29063836	BEST1 mediates Ca^{2+}-dependent function	Channel function in patient RPE

PMID	Title (abbrev.)	Role in this report
42010707	Allele-specific CRISPR suppression of BEST1	Rational therapy for dominant/GoF mutations
24143172	rAAV-mediated BEST1 transfer to RPE	Gene-delivery feasibility (canine)
42559932	Bestrophinopathies: natural history	Slow BCVA decline context (broader group)

Evidence source types used: human clinical/genetic (family and case series, registry), in vitro (minigene splicing, iPSC-RPE, MDCK), structural biology (cryo-EM), and model organism (canine, zebrafish).

Limitations and Knowledge Gaps

1. **No ADVIRC-specific functional/animal model.** Functional data (iPSC-RPE, canine cmr, zebrafish) overwhelmingly model BVMD/ARB loss-of-function or dominant-negative mutations. The ADVIRC gain-of-function splicing isoform mixture has **not** been directly characterized in a physiological channel-function assay, so "gain-of-function" is partly inferred from phenotype/imaging (Ni 2026) rather than from direct electrophysiology of the three co-assembled isoforms.
2. **Ultra-rarity limits epidemiology.** Prevalence is extrapolated from a single regional registry (Norway); global prevalence, geographic/ethnic distribution, and sex-specific data are essentially unknown.
3. **Variable expressivity is unexplained.** The same variant produces widely different severity within families, implying modifier genes or stochastic developmental factors that have not been identified.
4. **No disease-specific QoL instruments or natural-history cohort.** Prognostic estimates borrow from broader bestrophinopathy natural-history data.
5. **Mechanistic branch assignment partly inferred.** The precise reason splice-altering variants cause both developmental anomalies and degeneration (vs. degeneration alone) is not experimentally dissected.
6. **Variant spectrum is small.** Only a handful of ADVIRC variants are confirmed; genotype–phenotype correlation is underpowered.

Proposed Follow-up Experiments / Actions

1. **Direct functional characterization of ADVIRC isoform mixtures.** Express the three bestrophin-1 isoforms (wild-type + missense + in-frame deletion) at physiological stoichiometry in iPSC-RPE or heterologous pentameric reconstitution and perform patch-clamp + anion-flux assays to confirm gain-of-function at the channel level and define the biophysical change (e.g., altered Ca^{2+} sensitivity, altered Cl^- conductance).
2. **Generate an ADVIRC-specific model.** Create a knock-in mouse or zebrafish carrying an exonic splice variant (e.g., p.V86M-equivalent) to test whether both the developmental (nanophthalmos/anterior-segment) and degenerative (peripheral band) arms are recapitulated — currently a major gap.
3. **Allele-specific knockdown as therapy.** Extend allele-specific CRISPR/Cas9 or ASO suppression (PMID: 42010707) to ADVIRC splice variants in patient-derived iPSC-RPE and assess rescue of fluid flux, outer-segment handling, and channel function.
4. **Splice-correction approach.** Because the lesion is a splicing defect, test splice-switching ASOs or small molecules that restore normal exon inclusion, potentially eliminating the aberrant in-frame-deletion isoform.
5. **Modifier discovery.** Perform WGS + transcriptomics across multiple affected individuals sharing a variant but differing in severity to identify genetic/epigenetic modifiers of expressivity.
6. **Prospective natural-history registry.** Assemble a multicenter ADVIRC cohort with standardized ultra-widefield imaging, OCT, EOG/ffERG, and QoL instruments to define progression rates, prognostic biomarkers, and clinical-trial endpoints.
7. **Developmental-arm investigation.** Use ADVIRC iPSC-derived optic-vesicle/organoid systems to interrogate how the gain-of-function channel perturbs anterior-segment and axial-length development, linking the molecular defect to nanophthalmos and angle-closure glaucoma.

Report compiled from 7 confirmed findings and 52 reviewed papers over 5 investigation iterations. All mechanistic and clinical claims are cited to primary literature (PMID) with verbatim abstract quotes where available.